Chapter 1

Numerical sequences

Skills: search, model, represent, calculate, communicate

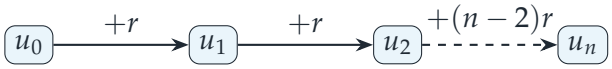
1.1 Reference notes

A sequence of numbers assigns to each natural number n a number u_n. The index n denotes an index; it does not must not be confused with the value u_n.

1.1.1 Arithmetic Sequence

Key ideas

A sequence (u_n) is **arithmetic** when there is a real number r such that, for all n, $u_{n+1} = u_n + r$. The number r is the common difference. If the first term is u_0, then $u_n = u_0 + nr$. If the first term is u_1, then $u_n = u_1 + (n-1)r$.



Three numbers x, y, z, written in this order, are three consecutive terms of an arithmetic sequence if $y - x = z - y$, i.e. if $y = \dfrac{x+z}{2}$. This last quantity is the arithmetic mean of x and z.

Method

To model an additive situation: 1) define u_n and specify the unit; 2) identify the first term; 3) express the constant increase or decrease as the common difference/ratio r ; 4) Write the term general case; 5) answer with a sentence in context.

Worked example

A shop sells 480 units in the first month, then 25 more units each month. With $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n-1)$. Thus $u_{12} = 755$: the business forecasts 755 units sold in the twelfth month.

1.1.2 Sum of arithmetic terms

For $n+1$ terms of u_0 to u_n,

$$u_0 + u_1 + \cdots + u_n = (n+1)\frac{u_0 + u_n}{2}.$$

The formula can be memorised as: number of terms multiplied by the mean of the first and last term .

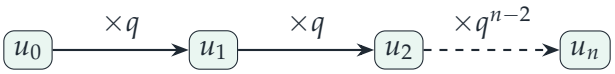
Common mistake

In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 9 or 12.

1.1.3 Geometric sequence

Key ideas

A sequence (u_n) is **geometric** when there is a real number q such that $u_{n+1} = qu_n$. If the first term is u_0, then $u_n = u_0 q^n$. If the first term is u_1, then $u_n = u_1 q^{n-1}$.



An increase of t % corresponds to $q = 1 + \dfrac{t}{100}$; a decrease of t % à $q = 1 - \dfrac{t}{100}$. Three positive terms x, y, z are consecutive in a geometric sequence if $y^2 = xz$; $y = \sqrt{xz}$ is their geometric mean.

For $q \neq 1$,

$$u_0 + u_1 + \cdots + u_n = u_0 \frac{1 - q^{n+1}}{1 - q}.$$

1.1.4 Algorithms: calculating a sum

```
def somme_carres(n):
    s = 0
    for k in range (1, n + 1):
        s = s + k**2
    return s
```

The accumulator `s` contains the partial sum; the counter `k` runs through the integers from 1 to n.

Session 1 — Arithmetic sequence: recognise and model

Numerical sequences

15/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Recognise an additive change.

· Determine the common difference/ratio.

· Define a sequence in a context.

Fluency · 6 to 8 minutes, without calculator

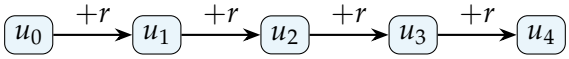
1. Write the multiplier for a 10% increase
2. Calculate u_4 if $u_0 = 14$ and $u_{n+1} = u_n + 3$. .....
3. How many terms of u_3 to u_{15} ? .....
4. Give the multiplier for a decrease of 8%
5. Give the common difference of $12, 12+4, 12+2\times 4, \ldots$ .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: recognise an additive change and determine the common ratio.

Investigation

An association has 320 members. It expects to gain 18 each year. We denote u_n the number after n years. Calculate u_0, u_1, u_2. Which operation takes one term to the next? Write a relationship between u_{n+1} and u_n, then predict u_5.



Written summary

A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a situation, an increase absolute change leads to an arithmetic sequence.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. The sequence (u_n) satisfies $u_0 = 110$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is 420 and decreases by 3% per period. Write v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

References in the textbook

Extensions: Calao, courses p. 18·21; exercises p. 24, n° 1·3.
session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 24, Exercise 5

For each of the sequences, specify whether it is arithmetic and, if so, indicate its common difference.

a. For every integer n : $u_n = 2(n-5) + 3n$.

b. For every integer n : $v_n = n^2 + n - 1$.

c. For every integer n : $w_n = -4 + 3n$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 125 and the rate or common ratio by 3. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 2 — Arithmetic sequence: general term

Numerical sequences

17/09/2026 • B : 1 h – C : 2 h

Objectives of the session

· Go from the recurrence relation to the general term.

· Calculate a distant term.

· Interpret the index.

Fluency · 6 to 8 minutes, without calculator

1. Simplify 3 $x + 5 - x + 2$. .....

2. How many terms of u_3 to u_{15} ? .....

3. Give the multiplier for a decrease of 8%

4. Give the common difference of $12, 12 + 5, 12 + 2 \times 5, \ldots$ .....

5. Calculate u_4 if $u_0 = 15$ and $u_{n+1} = u_n + 3$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: move from the recurrence to the general term and calculate a distant term.

Investigation

A shop has 1,250 active loyalty cards at index 0 and gains 75 per month. Without calculating all terms, determine the number at index n, then at index 18. Explain the role of 18 in the calculation.

Written summary

If (u_n) is arithmetic with common difference r, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, according to the index of the first term.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. The sequence (u_n) satisfies $u_0 = 120$ and $u_{n+1} = u_n + 4$. Write u_n then calculate u_{12}.

2. A quantity is 440 and decreases by 4% per period. Write v_n and calculate v_5.

3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

References in the textbook

Extensions: Calao, courses p. 18·21; exercises p. 24, n° 5·8.
session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 24, Exercise 6

Let (l_n) the arithmetic sequence of common difference 2 and first term: $l_0 = -1$.

a. Express l_n for every natural number n.

b. Calculate l_{35} and l_{64}.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 130 and the rate or common ratio by 4. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 3 — Sum of terms of an arithmetic sequence

Numerical sequences

22/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Count the terms of a sum.

· Apply the formula.

· Check the order of magnitude.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 120 .....

2. Give the multiplier for a decrease of 8%

3. Give the common difference of $12, 12+6, 12+2\times 6, \ldots$ .....

4. Calculate u_4 if $u_0 = 16$ and $u_{n+1} = u_n + 3$. .....

5. How many terms of u_3 to u_{15} ? .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: count the terms of a sum and apply the formula.

Investigation

A company pays a monthly bonus: 45 euros the first month, then 5 euros more each months. Calculate the total paid during the first twelve months in three ways: explicit sum, first-last pairing, formula.

Written summary

A sum of arithmetic terms is number of terms $\times \dfrac{\text{first} + \text{last}}{2}$. From u_p to u_q, there are $q - p + 1$ terms.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. The sequence (u_n) satisfies $u_0 = 130$ and $u_{n+1} = u_n + 5$. Write u_n then calculate u_{12}.

2. A quantity is 460 and decreases by 5% per period. Write v_n and calculate v_5.

3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

References in the textbook

Extensions: Calao, courses p. 20·21; exercises p. 24, n° 11-13, then p. 28, n° 43·44. reproduced in this session is identified below by its page and number; the reference allows also to find the figures and digital resources of the textbook.

Calao textbook · p. 24, Exercise 9

a. Calculate the sum of the first 500 non-zero integers $\sum_{i=1}^{500} i$.

b. Calculate the sum of the integers from 35 to 150: $\sum_{k=35}^{150} k$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 135 and the rate or common ratio by 5. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 4 — Geometric sequence: recognise and model

Numerical sequences

24/09/2026 • B : 1 h – C : 2 h

Objectives of the session

· Recognise a multiplicative change.

· Go from a rate to a multiplier.

· Compare arithmetic and geometric.

Fluency · 6 to 8 minutes, without calculator

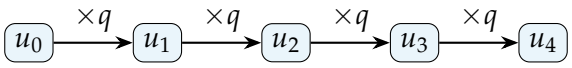
1. Write the multiplier for a 25% increase
2. Give the common difference of $12, 12+7, 12+2\times 7, \ldots$ .....
3. Calculate u_4 if $u_0 = 17$ and $u_{n+1} = u_n + 3$. .....
4. How many terms of u_3 to u_{15} ? .....
5. Give the multiplier for a decrease of 8%

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: recognise a multiplicative change and move from a rate to a multiplier.

Investigation

An online shop increases its number of visitors by 6 % per month, starting from 8,000 visitors. Calculate the first three values. Compare the differences, then the quotients. Identify the type of the sequence.



Written summary

A sequence is geometric when $\dfrac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). A relative change constant leads to a geometric sequence.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. The sequence (u_n) satisfies $u_0 = 140$ and $u_{n+1} = u_n + 6$. Write u_n then calculate u_{12}.
2. A quantity is 480 and decreases by 6% per period. Write v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a

difference or quotient.

References in the textbook

Extensions: Calao, courses p. 23·25; exercises p. 25, n° 14·18.
session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 25, Exercise 19

Let (l_n) the geometric sequence of common ratio 2 and first term: $l_0 = -1$.

a. Express l_n for every natural number n.

b. Calculate l_{12} and l_{20}.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 140 and the rate or common ratio by 6. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 5 — Geometric sequence: general term and sum

Numerical sequences

29/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Write the general term.

· Calculate a geometric sum.

· Interpret a cumulative change.

Fluency · 6 to 8 minutes, without calculator

1. Simplify $x + 8 - x + 2$. .....
2. Calculate u_4 if $u_0 = 18$ and $u_{n+1} = u_n + 3$. .....
3. How many terms of u_3 to u_{15} ? .....
4. Give the multiplier for a decrease of 8%
5. Give the common difference of $12, 12 + 8, 12 + 2 \times 8, \ldots$ .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: write the general term and calculate a sum geometric.

Investigation

A capital of 250,000 Ar is invested at 4 % per year, compound interest. Write the capital C_n after n years. Calculate C_6, then the sum $C_0 + \cdots + C_6$ and explain what this last sum mathematically represents and why it is not the final capital.

Written summary

If (u_n) is geometric with common ratio q, $u_n = u_0 q^n$ or $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. The sequence (u_n) satisfies $u_0 = 150$ and $u_{n+1} = u_n + 2$. Write u_n then calculate u_{12}.
2. A quantity is 500 and decreases by 2% per period. Write v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

References in the textbook

Extensions: Calao, courses p. 25·28; exercises p. 25, n° 19-25, and p. 28, n° 42·44. reproduced in this session is identified below by its page and number; the reference allows also to find the figures and digital resources of the textbook.

Calao textbook · p. 25, Exercise 22

Consider the sequence (u_n) geometric with common ratio 3 and first term: $u_0 = 5$.

a. Express u_n for every natural number n. Deduce u_7.

b. Calculate: $S = \sum_{i=0}^{7} u_i = u_0 + u_1 + \cdots + u_7$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 145 and the rate or common ratio by 7. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 6 — Review and formative assessment

Numerical sequences

01/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Choose the right model.

· Use the formulae.

· Analyse your errors.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 180 .....
2. How many terms of u_3 to u_{15} ? .....
3. Give the multiplier for a decrease of 8%
4. Give the common difference of $12, 12+9, 12+2\times 9, \ldots$ .....
5. Calculate u_4 if $u_0 = 19$ and $u_{n+1} = u_n + 3$. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: choose the right model and use the formulas.

Investigation

Classify eight situations into three categories: arithmetic, geometric, neither. For each situation, highlight the word or data that justifies the choice, then choose the appropriate formula.

Written summary

Before calculating, identify: type of sequence, first index, common difference or ratio, requested term or sum. The conclusion states the unit and the index.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. The sequence (u_n) satisfies $u_0 = 160$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is 520 and decreases by 3% per period. Write v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

References in the textbook

Extensions: Calao, p. 29, nos. 49 to 52. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 29, Exercise 49

For each statement, say whether it is true or false. Justify the answer.

1. A capital of 5,000 € is invested in a simple-interest account at an annual interest rate of 4%. This situation can be modelled by a geometric sequence with common ratio 1.04.
2. The number of sparrows in a region falls by 20% per year. An association decides to reintroduce 200 sparrows each year. This situation can be modelled by a sequence arithmetic with common difference 200.
3. The mass of household waste falls by 10% per year. This situation can be modelled by a geometric sequence of common ratio 1.1.
4. The production of a company increases weekly by 200 items produced. This situation can be modelled by an arithmetic sequence of common difference 200.
5. The number of subscribers of a YouTuber increases by 15% every month. This situation can be modelled by an arithmetic sequence of common difference 15.
6. The annual rent of a flat rises by 320 € every year. This situation can be modelled by an arithmetic sequence of common difference 320.
7. On 1 st January of a given year, Enzo deposits 1,000 € at compound interest in an account whose rate annual is 2 % and on 1 st January of each following year, he deposits 200 € on this account. This situation can be modelled by a geometric sequence with common ratio 1.02.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 150 and the rate or common ratio by 2. Write four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.